

Bank Account Management System using Classes and Objects

Experiment No: 04 | Date: 12/08/2026

Question

Implement a Bank Account Management System using Classes and Objects in Java, demonstrating deposit, withdrawal, balance checking, and displaying account details.

Steps

1. Create a BankAccount class with account number, account holder name, and balance as data members.
2. Create a constructor to initialize account details.
3. Create a deposit() method to add money.
4. Create a withdraw() method to withdraw money after checking available balance.
5. Create a checkBalance() method to display current balance.
6. Create a displayAccount() method to display account information.
7. Create a BankAccountManagement class with main().
8. Create a BankAccount object via the constructor.
9. Display a menu and let the user perform banking operations.
10. Repeat until the user selects Exit.

Algorithm

1. Start.
2. Input account number, account holder name, and initial balance.
3. Create a BankAccount object with these values.
4. Display menu: Deposit, Withdraw, Check Balance, Display Details, Exit.
5. Read user's choice.
6. If Deposit, read amount and call deposit().
7. If Withdraw, read amount and call withdraw() (checks sufficient balance).
8. If Check Balance, call checkBalance().
9. If Display Details, call displayAccount().
10. If Exit, print thank-you message and stop.
11. Repeat steps 4-10 until Exit is chosen.
12. Stop.

UML Class Diagram

```
+-----+
|           BankAccount           |
+-----+
| - accountNumber : String        |
| - accountHolder : String        |
| - balance : double              |
+-----+
| + BankAccount(...)              |
| + deposit(amount) : void        |
+-----+
```

```

| + withdraw(amount) : void      |
| + checkBalance() : void       |
| + displayAccount() : void     |
+-----+
      ^
      | creates
+-----+
|   BankAccountManagement      |
+-----+
| + main(args : String[]) : void |
+-----+

```

- = private + = public
BankAccountManagement creates and uses a BankAccount object.

Source Code

```

import java.util.Scanner;

class BankAccount {
    private String accountNumber;
    private String accountHolder;
    private double balance;

    BankAccount(String accountNumber, String accountHolder, double balance) {
        this.accountNumber = accountNumber;
        this.accountHolder = accountHolder;
        this.balance = balance;
    }

    void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Amount deposited successfully.");
        } else {
            System.out.println("Invalid amount.");
        }
    }

    void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.println("Amount withdrawn successfully.");
        } else {
            System.out.println("Insufficient balance or invalid amount.");
        }
    }

    void checkBalance() {
        System.out.println("Current Balance: Rs." + balance);
    }

    void displayAccount() {
        System.out.println();
        System.out.println("--- Account Details ---");
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder: " + accountHolder);
        System.out.println("Balance: Rs." + balance);
    }
}

public class BankAccountManagement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Account Number: ");
        String accNo = sc.nextLine();
    }
}

```

```

        System.out.print("Enter Account Holder Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Initial Balance: ");
        double balance = sc.nextDouble();

        BankAccount account = new BankAccount(accNo, name, balance);
        int choice;

        do {
            System.out.println();
            System.out.println("===== BANK ACCOUNT MANAGEMENT =====");
            System.out.println("1. Deposit");
            System.out.println("2. Withdraw");
            System.out.println("3. Check Balance");
            System.out.println("4. Display Account Details");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            switch (choice) {
                case 1:
                    System.out.print("Enter deposit amount: ");
                    account.deposit(sc.nextDouble());
                    break;
                case 2:
                    System.out.print("Enter withdrawal amount: ");
                    account.withdraw(sc.nextDouble());
                    break;
                case 3:
                    account.checkBalance();
                    break;
                case 4:
                    account.displayAccount();
                    break;
                case 5:
                    System.out.println("Thank you!");
                    break;
                default:
                    System.out.println("Invalid choice.");
            }
        } while (choice != 5);

        sc.close();
    }
}

```

Output Screen

```

Enter Account Number: 123456789
Enter Account Holder Name: Dinkar
Enter Initial Balance: 5000

===== BANK ACCOUNT MANAGEMENT =====
1. Deposit
2. Withdraw
3. Check Balance
4. Display Account Details
5. Exit
Enter your choice: 1

Enter deposit amount: 2000
Amount deposited successfully.

===== BANK ACCOUNT MANAGEMENT =====
1. Deposit
2. Withdraw

```

```
3. Check Balance
4. Display Account Details
5. Exit
Enter your choice: 2

Enter withdrawal amount: 1000
Amount withdrawn successfully.

===== BANK ACCOUNT MANAGEMENT =====
1. Deposit
2. Withdraw
3. Check Balance
4. Display Account Details
5. Exit
Enter your choice: 3

Current Balance: Rs.6000.0

Enter your choice: 4

--- Account Details ---
Account Number: 123456789
Account Holder: Dinkar
Balance: Rs.6000.0

Enter your choice: 5
Thank you!
```

Result

The Bank Account Management System was successfully implemented in Java using classes, objects, constructors, encapsulation, and methods.